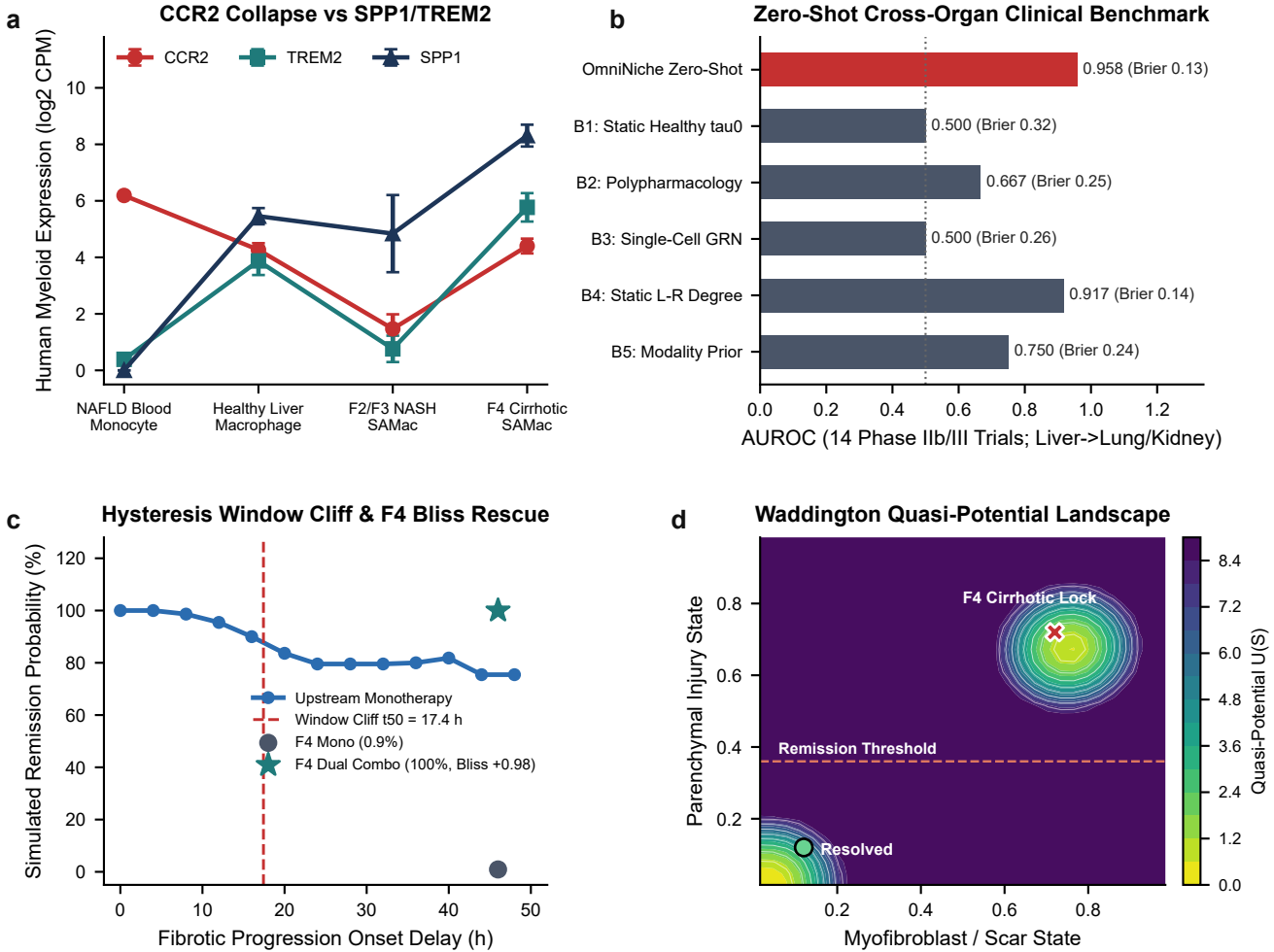

a
CCR2 Collapse vs SPP1/TREM2
CCR2
TREM2
SPP1
Human Myeloid Expression (log2 CPM)
0
2
4
6
8
10
NAFLD Blood Monocyte
Healthy Liver Macrophage
F2/F3 NASH SAMac
F4 Cirrhotic SAMac
b
Zero-Shot Cross-Organ Clinical Benchmark
OmniNiche Zero-Shot
0.958 (Brier 0.13)
B1: Static Healthy tau0
0.500 (Brier 0.32)
B2: Polypharmacology
0.667 (Brier 0.25)
B3: Single-Cell GRN
0.500 (Brier 0.26)
B4: Static L-R Degree
0.917 (Brier 0.14)
B5: Modality Prior
0.750 (Brier 0.24)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
AUROC (14 Phase IIb/III Trials; Liver->Lung/Kidney)
c
Hysteresis Window Cliff & F4 Bliss Rescue
Simulated Remission Probability (%)
0
20
40
60
80
100
120
Upstream Monotherapy
Window Cliff t50 = 17.4 h
F4 Mono (0.9%)
F4 Dual Combo (100%, Bliss +0.98)
0
10
20
30
40
50
Fibrotic Progression Onset Delay (h)
d
Waddington Quasi-Potential Landscape
F4 Cirrhotic Lock
Remission Threshold
Resolved
Parenchymal Injury State
0.2
0.4
0.6
0.8
0.2
0.4
0.6
0.8
Myofibroblast / Scar State
Quasi-Potential U(S)
0.0
1.2
2.4
3.6
4.8
6.0
7.2
8.4